

SECTION 15140-A

MECHANICAL EQUIPMENT INSTALLATION

PART 1 - GENERAL

1.1 SCOPE

This specification describes the requirements for mechanical equipment installation. This specification is intended to describe in general mechanical equipment installation and may be supplemented by specifications for specific equipment.

- A. Mechanical installation includes:
 - 1. assembly, if required and installation of all equipment either owner-supplied or contractor-supplied
 - 2. the unloading and storage of equipment
 - 3. moving equipment from storage to required location
 - 4. crane, rigging, temporary supports and protective materials and devices
 - 5. lubrication, testing and cleaning materials and devices
 - 6. bringing the equipment up to acceptance by owner
- B. "Equipment" in this specification means all process mechanical equipment including rotating and reciprocating machinery, apparatus, vessels, tanks, and includes drives, guards, piping, ducting, mounting devices and the like forming part of the equipment.

1.2 SUBMITTALS

- A. For contractor-supplied equipment submit follow owner's requested complete sets of vendor's drawings, installation, operating and maintenance instructions (including lubrication requirements) and parts lists four weeks prior to shipment of equipment.
- B. Retain one complete set of the vendor drawings and instructions for installation.
- C. Upon completion of the work, submit one complete set of all drawings, installation, operating and maintenance instructions and parts lists for all equipment, including "as-built" revisions.
- D. Supply all material and equipment, not furnished by owner, necessary to complete the work.

- E. Supply all miscellaneous steel, bolts, gaskets, welding equipment and materials, touch-up paint and the like not specifically shown on the project drawings or supplied with the equipment necessary to complete the installation and the work.

1. Steelwork

Conform to the requirements of CSA S16.1 for structural and miscellaneous steelwork. Conform to requirements of CSA W59 for welding. Use bolted connections for field assembly. Notify owner's Representative prior to welding to existing structural steel. Use ASTM A325 bolts, nuts and washers unless noted otherwise.

2. Lubricants

In accordance with manufacturer's recommendations unless otherwise directed by the owner's Representative.

PART 2 - PRODUCTS

2.1 GENERAL

The contract-supplied equipments are specified by the specific specifications for these equipments in other sections.

2.2 STRUCTURAL SUPPORTS

The contractor shall provide all supporting steel required for the installation of mechanical equipment and materials, whether or not it is specifically indicated or sized, including angles, channels, beams, etc. to suspend or floor support tanks and equipment.

PART 3 - EXECUTION

3.1 STORAGE AND HANDLING

Retrieve equipment from storage areas. Unload equipment from carrier vehicle and move to required location.

3.2 EXAMINATION

- A. Prior to commencing installation, thoroughly examine other work upon which mechanical equipment installation is dependent. Report deficiencies and propose adjustments to the owner's Representative. Obtain written authorization before proceeding.
- B. Examine equipment in storage to confirm setting dimensions in accordance with manufacturer's detail setting drawings.

3.3 PREPARATION

- A. Prepare surfaces and bases as required to receive mechanical equipment. Remove all grease, dirt, rust, oil and loose material from the underside of the equipment baseplates, concrete bases and steel supports.
- B. Ensure that the surface of the base has been properly finished to receive the equipment.
- C. Before placing equipment on steel supports ensure that contact surfaces with the equipment are properly prepared and cleaned.

3.4 INSTALLATION

- A. For owner supplied equipment, owner will furnish one complete set of vendor drawings and installation, operating and maintenance instruction.
- B. Assemble and install equipment in proper location, position and orientation in accordance with the manufacturer's instructions.
- C. Accurately mate and assemble component parts and subassemblies.
- D. Unless indicated otherwise level equipment baseplates to within 0.002 inches per foot and soleplates to within 0.001 inches per foot.
- E. Align equipment true to required lines and elevations.
- F. Install driver and driven equipment true and in line within acceptable tolerances.
- G. Connect piping and ductwork only after the equipment is in final position.
- H. Connect piping and ductwork so that no stress is transmitted to equipment.

- I. Properly install and secure guards per CSA Z432.
- J. Protect equipment against damage and ingress of foreign matter. At owner's discretion, if protection afforded is considered inadequate, open the equipment for internal examination. Repair, clean and re-assemble the equipment damaged by inadequate protection. Perform such work without additional payment.
- K. Service and lubricate equipment in accordance with the manufacturer's instruction and make ready for operation. Ensure grease points and lubrication devices operate correctly.
- L. Take special cleanliness precautions during hydraulic equipment installation. Properly install filters and strainers. Set drives to operate in proper direction.

3.5 INSPECTION AND TESTING

- A. Ensure equipment is properly installed and serviced prior to start up. Record initial alignment of shafts, coupling, bolt torques and the like. Notify the owner's Representative when equipment is ready to start up and commission.
- B. Do not start equipment until instructed by the owner's Representative.
- C. Assist and provide tools and test instruments as required to commission and performance test the equipment.

3.6 TOUCH-UP PAINT

- A. Repair damaged paint. Use paint compatible with and matching the original equipment paint in color and quality.

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